

Machine Learning 2

Aziz Temirkhanov

Ranking

What is ranking?

- Every time there is list of things, the list is **sorted**
- This sorting done by **ranking** the items
- Any search results in search engines, streaming services etc. are **ranked**
- There is also silent ranking. It is often done before the matchmaking algorithm

What is Ranking?

Google! ranking

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Ranking
A ranking is a relationship between a set of items, often recorded in a list, such that, for any two items, the first is either ranked higher than, ranked ...

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Просмотр Рекомендации Категории Способы игры Прочее Поиск по магазину

Ранняя игра Секция Фэнтези

Введите поисковое слово или метку Поиск Сортировать по релевантности Цена

Результаты по вашему запросу: 6.954. Истории: предметы (423) скрыты согласно вашим настройкам. Впрочем, ни один из них не появился бы на первой странице результатов поиска.

Игра	Дата выпуска	Цена
DOTA 2	9 июл. 2013 г.	Бесплатно
Black Desert	28 июл. 2019 г.	350 руб
Clair Obscur: Expedition 33	24 апр. 2025 г.	3499 руб
ELDEN RING	25 фев. 2022 г.	3599 руб
Black Myth: Wukong	20 авг. 2024 г.	3599 руб
Albion Online	16 мар. 2018 г.	Бесплатно
ARK: Survival Ascended	26 окт. 2023 г.	4230 руб
ENSHROUDED	24 мар. 2024 г.	1399 руб
Torchlight: Infinite	10 мар. 2023 г.	Бесплатно
Warhammer 40,000: Rogue Trader	7 авг. 2023 г.	1599 руб
ELDEN RING Shadow of the Erdree	21 июл. 2024 г.	2399 руб
Last Epoch	21 фев. 2024 г.	1475 руб
Path of Exile	23 окт. 2013 г.	Бесплатно

Язык: русский, английский, китайский (упр.), китайский (трад.), японский, корейский, тайский, арабский, болгарский, чешский, датский, немецкий, испанский — Испания, испанский — Латинская Америка, греческий, французский, найти другие языки

Управление языковыми настройками

Предпочтения: Не показывать рекламу, Не показывать купленные, Не показывать начисленные

Метки: Ранняя игра, Экшен, Фэнтези

Для одного игрока 4.468, Преположение 4.186

Problem setup

- Documents D , queries Q , candidates D_q per query q
- Goal: sort D_q by decreasing relevance to q
- Typical pipeline: candidate generation → final ranking

Examples

- Web search: sort indexed pages by usefulness to a query
- Synonym search: rank words by semantic similarity to a given word
- Recommenders: rank items for a user; user history plays the role of the query

Metrics

Metrics

Assume, we trained a ranking model. Usually it's done by optimizing some function

$$a_{\theta}(q, d)$$

Then, results of query $q \in Q$ is a ranking/sorting of D_q with decreasing $a_{\theta}(q, d)$

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WLOG, assume we solving ranking for WEB-search. Take Q_t — queries test set.

$T_K(q)$ — first K results for q ; $d_q^{(k)}$ — k-th results in a ranking. Then: $T_K(q) = \{d_q^{(1)}, \dots, d_q^{(k)}\}$

Binary Relevance

- Assessor-provided labels for (query, document) pairs: $y(q, d)$
- Binary labels: {0,1} or graded labels: e.g., NotRel ... Vital
- Datasets combine queries, candidate docs, and relevance labels

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First, let us introduce metrics for the $y(q, d) \in \{0, 1\}$.

Precision and Recall

$$\textit{Precision@}K = \frac{\sum_{d \in T_K(q)} I\{y(q, d) = 1\}}{K}$$

$$\textit{Recall@}K = \frac{\sum_{d \in T_K(q)} I\{y(q, d) = 1\}}{\min(K, \sum_{d \in D} I\{y(q, d) = 1\})}$$

mAP

$$AP(q) = \frac{1}{K} \sum_{k=1}^K Precision@k \cdot y(q, d_q^{(k)})$$

$$MAP(q) = \frac{1}{|Q_t|} \sum_{q \in Q_t} AP(q)$$

Mean Reciprocal Rank

$$MRR = \frac{1}{|Q|} \sum_{q \in Q} \min \left(i \mid y(q, d_q^{(i)}) = 1 \right)^{-1}$$

Relevance of k-th document as $R_k = y(q, d_q^{(k)})$

$$MRR = \frac{1}{|Q|} \sum_{q \in Q} \sum_i \prod_{k=1}^{i-1} (1 - R_k) \cdot R_i \cdot \frac{1}{i}$$

Expected Reciprocal Rank

Let $R_i = P(q, d_q^{(i)})$ be a probability that document $d_q^{(i)}$ is relevant.

If y_i is a relevance label, and Y_{max} it's maximum value, $R_i = \frac{2^{y_i}}{2^{Y_{max}}}$

Since user stopped at i he hasn't solved his problem at previous steps, thus consider those event independent.

$$ERR(q) = \sum_i \prod_{k=1}^{i-1} (1 - R_k) \cdot R_i \cdot \frac{1}{i}$$

$$ERR = \frac{1}{|Q|} \sum_{q \in Q} \sum_i \prod_{k=1}^{i-1} (1 - R_k) \cdot R_i \cdot \frac{1}{i}$$

pFound

$y(q, d) \in \mathbb{Y} = \{\text{Not Rel}, \text{Rel-}, \text{Rel+}, \text{Useful}, \text{Vital}\}$

$P_q(d) = \{0, 0.07, 0.14, 0.41, 0.61\}$

$$pFound(q) = \sum_{i=1}^n P_i \cdot P_q(d_q^{(i)})$$

nDCG

Define Discounted Cumulative Gain metric: $DCG_n(q) = \sum_{i=1}^n G_q(d_q^{(i)}) \cdot D(i)$

Relevant document bring some gain based on its relevance. User rarely scroll down to the lowest documents, thus their gain is lower (gain decreases).

$$G_q(d) = y(q, d); D(i) = \frac{1}{i+1}$$

$$nDCG(q) = \frac{DCG(q)}{\max DCG(q)}$$

Learn to Rank

Learn to Rank

- Pointwise: predict labels/scores per (q,d) independently
- Pairwise: compare pairs (or triples) to infer relative order
- Listwise: optimize a list-level objective directly

Pointwise

Assume we have Q , and for each quarry $q \in Q$ we have a set of documents we have to rank D_q .

Set of all possible assessments — $\mathbb{Y}$, one assessment $y(q, d)$

Then we can train any linear model, boosting or NN by minimizing MSE:

$$\sum_{q \in Q} \sum_{d \in D_q} (\hat{y}(q, d) - y(q, d))^2 \rightarrow \min$$

Pairwise

- Train on tuples (q, d_1, d_2) where $d_1 > d_2$ for query q
- Simple classifier as comparator may break transitivity
- Need structures that ensure consistent global ordering

Pairwise. RankingSVM

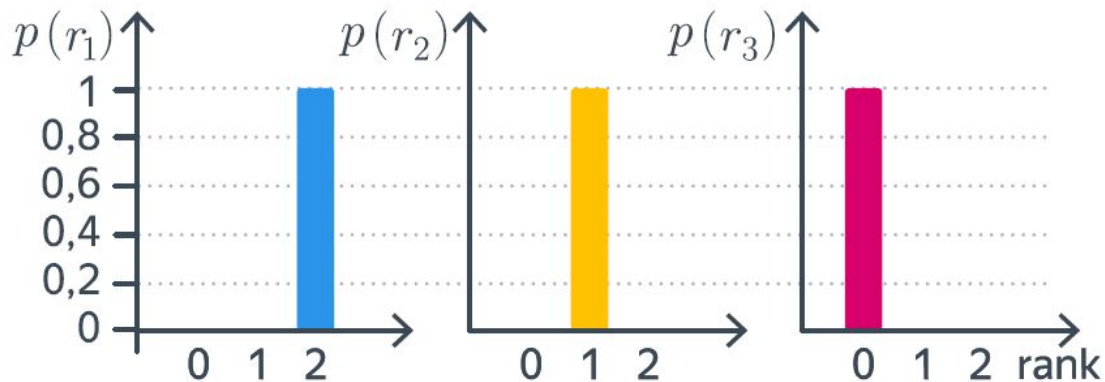
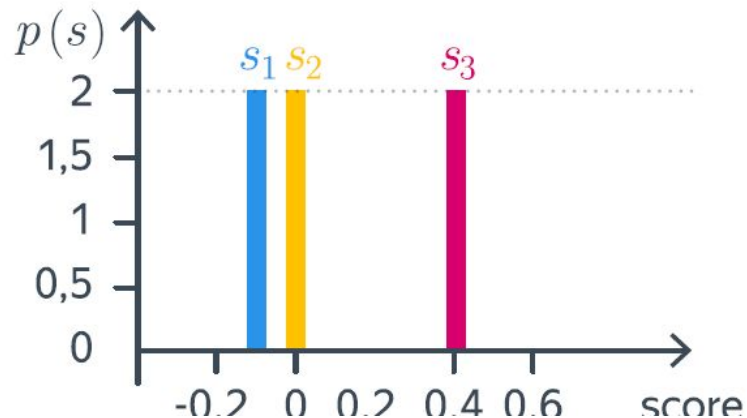
Find θ : $\forall q \in Q \forall d_i, d_j : d_i \prec d_j$ for which $\langle x_i, \theta \rangle < \langle x_j, \theta \rangle$ holds true

Train linear model to find $\langle x_i - x_j, \theta \rangle$. If you take SMV — you'll get RankingSVM.

To compare two documents by query you have to compare $\langle x_i - x_j, \theta \rangle$ with zero

Listwise. SofrRank

Assume we have d_1 , d_2 , and d_3 for query q . Ranking model yielded s_1 , s_2 , and s_3



Listwise. SofrRank

Assume that ranking evaluation d_i has a distribution $\mathcal{N}(s_i, \sigma^2)$

$$G_{soft} = \frac{1}{G_{max}} \sum_{j=1}^N g(d_j) \mathbb{E} D(j)$$

